Projected queries: checked native support and remaining IC work

ECDLP research programme

2026-10-26

This report covers native query and supplied-relation verification. It contains no new projected-curve panel, timing comparison, complete IC price or measured crossover. The original subgroup and cofactor receipts remain unchanged.

Conditional query law

Assume a correct finite abelian curve group of certified order hr, a prime r, $\gcd(h, r) = 1$, and a generator G of exact order r. Let $H = \langle G \rangle$ and $K = \ker[h]$. Then addition identifies $H \times K$ with the curve group. For a public target $T = [d]G$, independent uniform $k \in \mathbb{Z}/r$ and $U \in K$ give a uniform full-curve query

$$R = T + [k]G + U.$$

The helper accepts indices; it supplies no random sampler. The earlier draft with $U = O$ samples only H, so it cannot justify full-curve relation yield. Kernel randomization corrects that design before a projected-IC experiment.

For a verified full-curve relation $\sum_i \epsilon_i P_i = R$, each supplied log must satisfy $[L_i]G = [h]P_i$. Then

$$S = \sum_i \epsilon_i L_i = h(d + k) \pmod{r}, \qquad d = h^{-1}S - k \pmod{r}.$$

Native recovery checks every projected log, the signed full-curve relation and finally $[d]G = T$. Identity hits, zero projections and repeated or negative terms remain explicit. Neither equation discovers a relation or computes base logs.

Native implementation and verification

The original cyclic constructor checks $h \leq 4096$. The subsequent supplied-kernel constructor accepts an ordered complete census with $h \leq 65536$, including noncyclic kernels. It checks generator/order consistency, every element's membership and annihilation, exact length, uniqueness and identity. It retains caller order as the index map. Completeness is conditional on the certified ambient order and correct backend; the helper does not create either certificate.

The bound includes 4232. Its corresponding source test uses an abstract cyclic group, not the original E0 curve. No E0 kernel or new qualified-cell certificate has been produced. The noncyclic fixture is $\mathbb{Z}/7 \times (\mathbb{Z}/2)^2$; all targets and all query indices are checked. A separate 7-bit binary-curve fixture uses seed 202610260001 and independently enumerated polynomial-basis points/logs; its generator uses at most 16 construction attempts. This is a source test domain, not a measured IC rung or replacement for the n=22/25 panels.

Check	Recorded outcome and scope
Original helper source `0670ec6fa`	Eight Cargo tests pass; all earlier sparse failures retained
Original default Clippy	Exit 0; seven unchanged library warnings, zero own-helper diagnostics
Supplied-kernel source gate	Source `84ddcb9b0`: 11 Cargo tests and 10 bare-core tests pass; default lint exits 0 with seven unchanged library warnings
Existing Lean query dependencies	Five pass, zero trusted additions; general group bridge open
Exact residual certificate and proof	`IDC1h08a8b0da9bbd7a55`: separate native check passes; one exact ring lemma passes, zero trusted additions

The supplied-kernel gates ran at `84ddcb9b0`; earlier checks and the certificate retain their own recorded revisions. Integration against a newer repository base requires its own check. These are correctness gates on a shared Linux workspace, with no admitted performance isolation, runtime metric or statistical effect. The 7-bit fixture and abstract groups establish their finite checks only. All construction, field, membership, kernel-storage and query work would need pricing in a full instrument; a group-call ledger alone is incomplete.

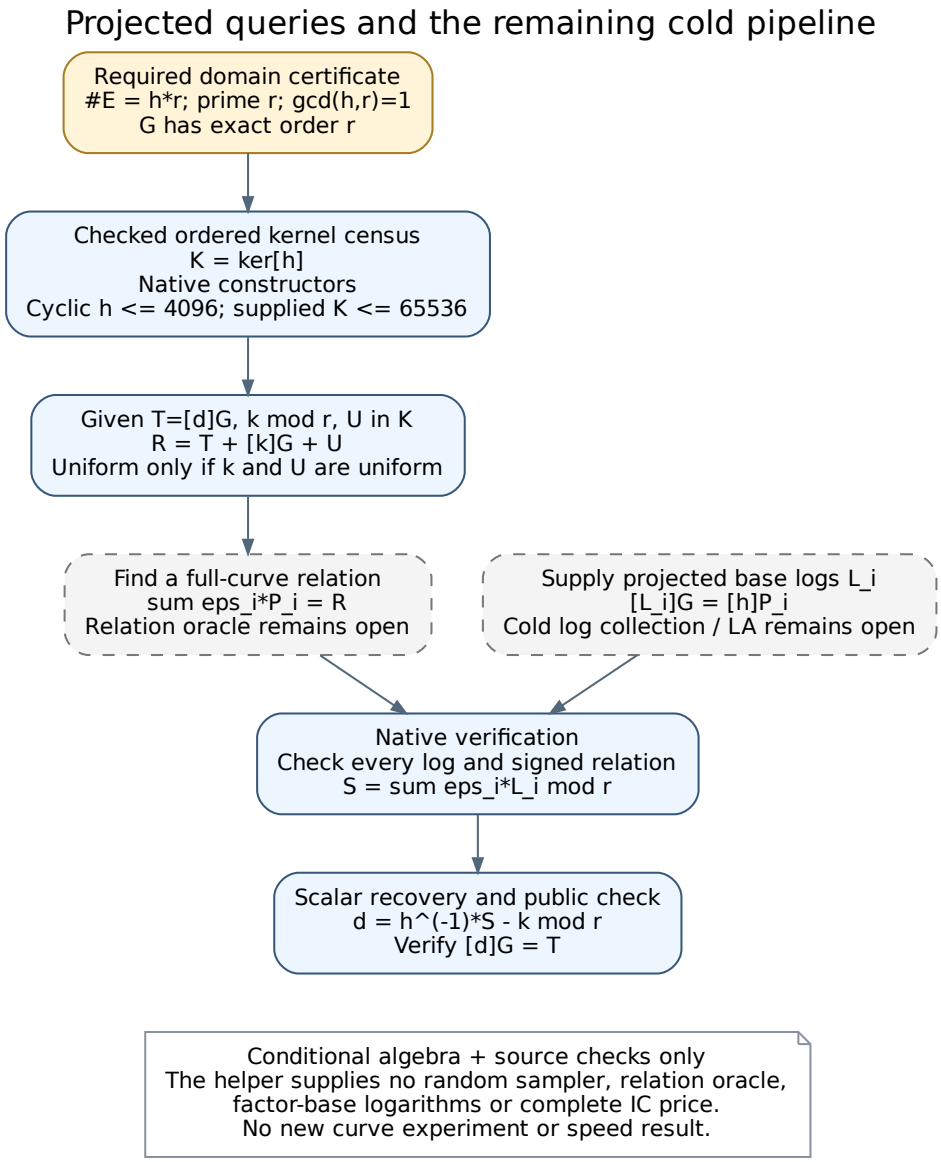


Figure 1. Solid steps are checked native support; dashed boxes remain missing. Source: query-law.dot and the native example. No runtime or relation-yield measurement is plotted.

The supplied kernel closes an interface limitation, conditional on an actual complete census. The relation oracle, projected factor-base logs and mod-r linear algebra are separate missing components. A useful projected relation pipeline must also handle signed deduplication, target solving, every failed query and final public-scalar verification.

Proof and certificate boundary

The five existing Lean declarations prove explicit additive-projection, commutative-ring recovery/setup/identity-hit, and supplied-coordinate uniqueness dependencies. They do not construct the finite-group decomposition, certify curve cardinality or prove a concrete sampler uniform.

The new residual statement is the unconditional polynomial identity

$$h_{\text{supplied}}^{-1}\big(h(d+k)\big) - k - d = (h_{\text{supplied}}^{-1}h - 1)(d+k).$$

Here h_{supplied}^{-1} is just a variable; vanishing additionally requires it to be an actual inverse. The identity-certificate v1 checker evaluates a fixed modulus $2^{61}-1$ at 32 statement-derived points, with a separate checker using 256 extra domain-separated points. Its Lean pointer must be checked separately; fixed-modulus evaluations alone are not a proof over every ring. The polynomial has degree 3 and coefficients of magnitude 1. The general-group cardinality/uniformity bridge remains open.

Sources, receipts and next test

The implementation follows the study's cofactor audit, PROTOCOL.md and lead atlas entry LEAD-ICB-cofactor-torsion-quotients. SEC1v2 sections 3.1.2, 3.1.2.2 and 3.3.2 were read for binary-curve/order/cofactor conventions and validation: https://www.secg.org/sec1-v2.pdf. Download SHA256:

`54687189af3de645756a30a801b596fd14c6566992c13bfd58eede399f324a89`.

Gaudry 2009, DOI 10.1016/j.jsc.2008.08.005, was located in official HAL metadata, but its PDF returned 403; it is not claimed as read. No novelty or asymptotic conclusion rests on that inaccessible source.

Repository paths below use this study root:

`research/ic_algebraic_crossover_20261023`.

- Implementation/source limitations.
- Native source: `examples/ic_projected_query_law.rs` and its `core.rs` module; separate certificate example `examples/ic_projected_recovery_identity.rs`.
- Original exact gates on source `0670ec6fa`; failed attempts 1 and 2 remain alongside them.
- New gates: supplied kernel and residual identity.
- Proofs: `proofs/ProjectedQueryLaw.lean` and `proofs/ProjectedRecoveryIdentity.lean`.

Next, construct and independently verify complete order/kernel receipts for the actual qualified models, implement a uniform bounded sampler and complete cold projected relation/log/LA/recovery instrument, then freeze a controlled full-cost comparison against the current IC baseline and strongest matched rho. Preserve all failures, original workloads and total accounting. No additional VM launched; the total additional-cloud cap remains USD100 including replay, with workspace billing unmeasured. Larger 64/256 GiB requirements and complete wide quartic adapters remain outstanding.

Canonical scoreboard, progress timeline and performance figures are unchanged: these source checks introduce no admitted runtime, yield or crossover value.